

Zhaohan Xi

[\[Google Scholar\]](#)

Email: zxi1@binghamton.edu

Phone: +1 (484) 456-0211

Website: zhaohan-xi.github.io

G06, Engineering Building
Binghamton University (SUNY)
Vestal, NY 13850, USA

Research Interests

My current research focuses on developing cross-disciplinary AI solutions for *Decision Support* and *High-stakes* practices through *Large Language Models*, fostering responsible, robust, and resilient agentic systems, and enhanced by innovations in *Natural Language Processing*, *Knowledge Integration*, *Prompt Engineering*, and *Automatic Machine Learning* to bridge AI capabilities with trustworthy real-world deployment.

Keywords: Large Language Models (LLMs); Agentic AI; Responsible & Ethical AI; AI Security

Professional Appointment

2024 – on **Assistant Professor, Tenure-Track**
Binghamton University, the State University of New York (SUNY)
School of Computing
Binghamton, NY, USA

Education

2020 – 2024 **Ph.D.**
Pennsylvania State University, PA
College of Information Science and Technology
Informatics, Security and Privacy

2023 – 2024 **Visiting Scholar**
Stony Brook University, NY
College of Engineering and Applied Sciences
Department of Computer Science

2016 – 2018 **M.S.**
Lehigh University, PA
P.C. Rossin College of Engineering and Applied Science
Department of Computer Science and Engineering

2012 – 2016 **B.S.**
Nanjing University of Aeronautics and Astronautics, China
College of Electronic Engineering and Automation
Department of Automation

Publications

- ICML'26 The Value of Variance: Mitigating Debate Collapse in Multi-Agent Systems via Uncertainty-Driven Policy Optimization 
 Luoxi Tang, Yuqiao Meng, Joseph Costa, Yingxue Zhang, Muchao Ye, **Zhaohan Xi**
In Proceedings of the Forty-Third International Conference on Machine Learning (ICML 2026, Spotlight, 2.2% acceptance rate).
- ICML'26 Benchmarking LLMs in an Embodied Environment for Blue Team Threat Hunting. 
 Xiaoqun Liu, Feiyang Yu, Xi Li, Guanhua Yan, Ping Yang, **Zhaohan Xi**
In Proceedings of the Forty-Third International Conference on Machine Learning (ICML 2026).
- ICML'26 Small Agent Group is the Future of Digital Health 
 Yuqiao Meng, Luoxi Tang, Dazheng Zhang, Rafael Brens, Elvys J. Romero, Nancy Guo, Safa elkefi, **Zhaohan Xi**
In Proceedings of the Forty-Third International Conference on Machine Learning (ICML 2026).
- ICLR'26 On the Eligibility of LLMs for Counterfactual Reasoning: A Decompositional Study. 
 Shuai Yang*, Qi Yang, Luoxi Tang*, Jeremy Blackburn, **Zhaohan Xi**
In Proceedings of the 14th International Conference on Learning Representations (ICLR 2026).
- NPJ'26 TheraMind: A Multi-LLM Ensemble for Accelerating Drug Repurposing in Lung Cancer via Case Report Mining 
 Vrushket More, Lyra Lu , Shayna Kaye, Atharva Raje, Ryan Ginsburg, Zeyu Ding, **Zhaohan Xi**, Nancy L. Guo
NPJ Precision Oncology.
- IISE'26 HopeLink: A Large Language Model-Powered Clinical Information Assistant Supporting Cancer Patients' Online Information Seeking. 
 Vijayaraghavan, Balaji Jegadeesh, Yuqiao Meng, Srii Rohit Prakash, Mohamed Amine Kefi, Sriharshini Kalvakuntla, Safa Elkefi, **Zhaohan Xi**
In Proceedings of the 2026 IISE Annual Conference & Expo. Institute of Industrial and Systems Engineers.
- EMNLP'25 Data to Defense: The Role of Curation in Customizing LLMs Against Jailbreaking Attacks 
 Xiaoqun Liu, Jiacheng Liang, Luoxi Tang, Muchao Ye, Weicheng Ma, **Zhaohan Xi**
In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025).
- USENIX'24 On the Difficulty of Defending Contrastive Learning against Backdoor Attacks. 
 Changjiang Li, Ren Pang, Bochuan Cao, **Zhaohan Xi**, Jinghui Chen, Shouling Ji, Ting Wang.
In Proceedings of the 33rd USENIX Security Symposium (USENIX Security 2024).
- NAACL'24 PromptFix: Few-shot Backdoor Removal via Adversarial Prompt Tuning. 
 Tianrong Zhang, **Zhaohan Xi**, Ting Wang, Prasenjit Mitra and Jinghui Chen.
In Proceedings of the Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2024).
- USENIX'23 On the Security Risks of Knowledge Graph Reasoning. 
Zhaohan Xi, Tianyu Du, Changjiang Li, Ren Pang, Shouling Ji, Xiapu Luo, Xusheng Xiao, Fenglong Ma and Ting Wang.
In Proceedings of the 32nd USENIX Security Symposium (USENIX Security 2023).

- NeurIPS'23 Defending Pre-trained Language Models as Few-shot Learners against Backdoor Attacks. 
Zhaohan Xi, Tianyu Du, Changjiang Li, Ren Pang, Shouling Ji, Jinghui Chen, Fenglong Ma and Ting Wang.
In Proceedings of the 37th Conference on Neural Information Processing Systems (NeurIPS 2023) .
- ICLR'23 Neural Architectural Backdoors. 
Ren Pang, Changjiang Li, **Zhaohan Xi**, Shouling Ji and Ting Wang.
In Proceedings of the 11th International Conference on Learning Representations (ICLR 2023).
- ICCV'23 An Embarrassingly Simple Backdoor Attack on Self-supervised Learning. 
Changjiang Li, Ren Pang, **Zhaohan Xi**, Tianyu Du, Shouling Ji, Ting Wang and Yuan Yao.
In Proceedings of the IEEE International Conference on Computer Vision (ICCV 2023).
- USENIX'22 On the Security Risks of AutoML. 
Ren Pang, **Zhaohan Xi**, Shouling Ji, Xiapu Luo and Ting Wang.
In Proceedings of the 31st USENIX Security Symposium (USENIX Security 2022).
- USENIX'22 Seeing is Living? Rethinking the Security of Facial Liveness Verification in the Deepfake Era. 
Changjiang Li, Li Wang, Shouling Ji, Xuhong Zhang, **Zhaohan Xi**, Shanqing Guo and Ting Wang.
In Proceedings of the 31st USENIX Security Symposium (USENIX Security 2022).
- Euro SP'22 TROJANZOO: Towards unified, holistic, and practical evaluation of neural backdoors. 
Ren Pang, Zheng Zhang, Xiangshan Gao, **Zhaohan Xi**, Shouling Ji, Peng Cheng, Xiapu Luo, Ting Wang.
In Proceedings of the 7th IEEE European Symposium on Security and Privacy (EuroS&P 2022).
- USENIX'21 Graph backdoor. 
Zhaohan Xi, Ren Pang, Shouling Ji and Ting Wang.
In Proceedings of the 30th USENIX Security Symposium (USENIX Security 2021).
- USENIX'18 Towards a Secure Zero-rating Framework with Three Parties 
Zhiheng Liu, Zhen Zhang, Yinzhi Cao, **Zhaohan Xi**, Shihao Jing and Humberto La Roche.
In Proceedings of the 27th USENIX Security Symposium (USENIX Security 2018).

Under Review

- 2026 From Test-taking to Cognitive Scaffolding: A Pedagogical Diagnostic Benchmark for LLMs on English Standardized Tests
Luoxi Tang*, Tharunya Sundar*, Shuai Yang*, Ankita Patra*, Lakshmi Manohar Chippada*, Jiqian Zhao, Yi Li, Riteng Zhang, Tunan Zhao, Ting Yang, Yuqiao Meng*, Weicheng Ma, **Zhaohan Xi**
Under Review by EMNLP'26.

- 2026 Language Model Augmented Decision Transformer for Multi-Task Offline Reinforcement Learning
Xinbo Zhao, Xin Zhang, Lei Le, **Zhaohan Xi**, Yu Yang, Yingxue Zhang
Under Review by ICML 2026.
- 2026 ER-MIA: Black-Box Adversarial Memory Injection Attacks on Long-Term Memory-Augmented Large Language Models
Mitchell Piehl, **Zhaohan Xi**, Zuobin Xiong, Pan He, Muchao Ye
Under Review by NeurIPS 2026.
- 2026 Polar: Automating Cyber Threat Prioritization through LLM-Powered Assessment
Luoxi Tang, Yuqiao Meng, Ankita Patra, Weicheng Ma, Muchao Ye, **Zhaohan Xi**
Under Review by ICML 2026.

Professional Experience

Industry Experience

- 2024 **Sony Research Internship** | Sony AI America Inc
Mentors: Jihun Choi (Senior Research Scientist, Sony AI @ Japan)
Keywords: *LLMs, Prompt Engineering, Retrieval Augment Generation, Knowledge Graph*
- 2024 **Research Scientist** | ZBeats Inc
Mentors: Peng Zhang (Co-Founder, CTO)
Keywords: *LLM-as-Agent, Cardiological AI, Academic Writing, Clinical AI Infrastructure*
- 2023 **Data Scientist Internship** | Microsoft Turing Team, Bing Chat
Mentors: [Sukanya Patil](#), [Ishmam Zabir](#), [Tatsuro Oya](#). Manager: [Saksham Singhal](#)
Highlights:
 - Engaged in projects related to improving *Bing Chat*'s responsibility.
 - Focused on the development of *Prompt Engineering* solutions for responsible AI.
 - Worked with *GPT-4* (Large Language Models) to apply prompts effectively.
 - Explored *AutoML* techniques for the automated generation of prompts.
 - Implemented *Knowledge Distillation* to compress models and reduce overheads.**Keywords:** *LLMs, GPT-4, Prompt Engineering, Responsible AI, AutoML, Knowledge Distillation*
- 2022 **Ph.D. Software Engineering Internship** | Uber AI
Mentors: [Raajay Viswanathan](#). Manager: [Xu Ning](#)
Highlights:
 - Worked on *ML infrastructure*, a product developed by the Uber AI team.
 - Specialized in *AutoML*, focusing on tabular data used in Uber Eats & Rides.
 - Contributed to the development of two key functions within the AutoML Infra: *neural architecture search* and *feature selection*.
 - Conducted *data synthesis* to generate test cases for evaluating AutoML functionalities.**Keywords:** *AutoML, Neural Architecture Search, Data synthesis, ML Infrastructure*

Teaching

Instructor

- 2025 CS 436/536 “Introduction of Machine Learning”
School of Computing, *Binghamton University (SUNY)*.
- 2024,2025 INFO 561X “Large Language Model: Foundations and Applications”
School of Computing, *Binghamton University (SUNY)*.

Teaching Assistant

- 2022 Teaching Assistant of “Malware Analytics (Cyber-366).”
College of Information Science and Technology, *Pennsylvania State University*.

Invited Talk

Conference Presentation

- 2023 Presentation at NeurIPS 2023, *New Orleans, LA*.
- 2023 Presentation at USENIX Security Symposium 2023, *Anaheim, CA*.

Guest Lecture

- 2025 Presentation at Data Salon, Data Science Transdisciplinary Area of Excellence, Binghamton University, *Binghamton, NY*.
- 2024 Presentation at Weekly Seminar, School of Computing, Binghamton University, *Binghamton, NY*.
- 2022, 2023 Guest Lectures on the Course “Malware Analytics (Cyber-366)”, *Pennsylvania State University*.

Academic Services

Funding Agency Panelist

- 2026 NSF/CISE/Future Core: Robust Intelligence (RI).
- 2026 NSF/CISE/Future Core: Information Integration and Informatics (III).
- 2025 NSF/CISE/IIS: Robust Intelligence (RI).
- 2025 NSF Security, Privacy, and Trust in Cyberspace (SaTC).
- 2025 NSF Computer and Information Science and Engineering Research Initiation Initiative (CRII).
- 2024 NSF/CISE/IIS: Information Integration and Informatics (III).

Area Chair (AC), Program Committee (PC) or Reviewer

- 2026 Area Chair of the Annual Meeting of the Association for Computational Linguistics – Annual Rolling Review (ACL-ARR).
- 2026 Reviewer of The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

2025=Current Reviewer of The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP).

2025=Current Reviewer of The International Conference on Learning Representations (ICLR).

2025 Reviewer of the IEEE Transactions on Information Forensics & Security.

2025=Current Reviewer of Annual Meeting of the Association for Computational Linguistics – Annual Rolling Review (ACL-ARR).

2023 – 2025 Program Committee of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD).

2023=Current Reviewer of the Annual Meeting of the Association for Computational Linguistics (ACL).

2022 – 2023 Reviewer of the ACM International Conference on Information and Knowledge Management (CIKM).

2022 – 2023 Reviewer of the The Journal of Computer Security.